ECE343 Spring 2025

Objective 5.2:
Gauss' Law



UNITED STATES
AIR FORCE
ACADEMY

Lesson	Topics	Reading	Notes
24	BVP (Capacitor, Coax), Electric Potential	5.16	LO 5.3
25	Electric Boundary Conditions	5.18-5.22	LO 5.4
26	BC's/Capacitance	5.23-5.24	
27	Electrostatic Lab	Lab packet	
Chapter 6: Steady Current & Conductivity			
28	Current density, Conductivity, Resistance	Ch. 6	
Chapter 7: Magnetostatics			
29	Magnetic Fields, Gauss' Law for Magnetism	7.1-7.3	LO 7.1-7.2 **GR 2**
30	Ampere's Law	7.4-7.5, 7.9	LO 7.3
31	Ampere's Law Applied	7.6-7.8	LO 7.3
32	Magnetic Boundary Conditions	7.10-7.14	LO 7.4
33	Magnetostatics Lab	Lab packet	
Chapter 8: Time-Varying Fields			
34	Faraday's Law, Displacement Current, Maxwell Eqns	8.3, 8.8, 8.9	LO 8.1
35	Faraday's Law, Displacement Current, Maxwell Eqns	8.3, 8.8, 8.9	LO 8.2-8.3
Chapter 9: Plane Waves in Lossless Media			
36	Maxwell's Equations in lossless, source-free region	9.1-9.2	LO 9.1
37	Maxwell's Equations in lossless, source-free region	9.1-9.2	LO 9.1
38	Uniform Plane Waves & Polarization	9.4-9.6	LO 9.2-9.3
39	Normal Incidence	Vol 2, 5.6-5.7	LO 9.4-9.5
40	Oblique Incidence	Vol 2, 5.6-5.7	LO 9.5
41	Oblique Incidence	Vol 2, 5.6-5.7	LO 9.5
42	Incidence Lab	TBD	LO 9.5

Ch 5 Learning Objectives

5.1 I can apply Coloumb's Law to use direct integration to calculate the electric field due to charge density in varoius geometries, including rectangular, cylindrical, and spherical coordinate systems.

5.2 I can apply the integral and differential forms of Gauss' Law to calculate the electric field due to charge density in various geometries, including rectangular, cylindrical, and spherical coordinate systems.

5.3 I can calculate the electric field from the scalar electric potential in a variety of geometries.

5.4 I can apply the electric field boundary conditions to calculate electric fields on both sides of a material boundary.

Gauss' Law

Definition

The flux of the electric charge through a **closed** surface is equal to the enclosed charge.

Integral & Differential (Point) Forms

$$\oiint_S \vec{D} \cdot d\vec{S} = Q_{\text{enclosed}} \qquad \nabla \cdot \vec{D} = \rho_V$$

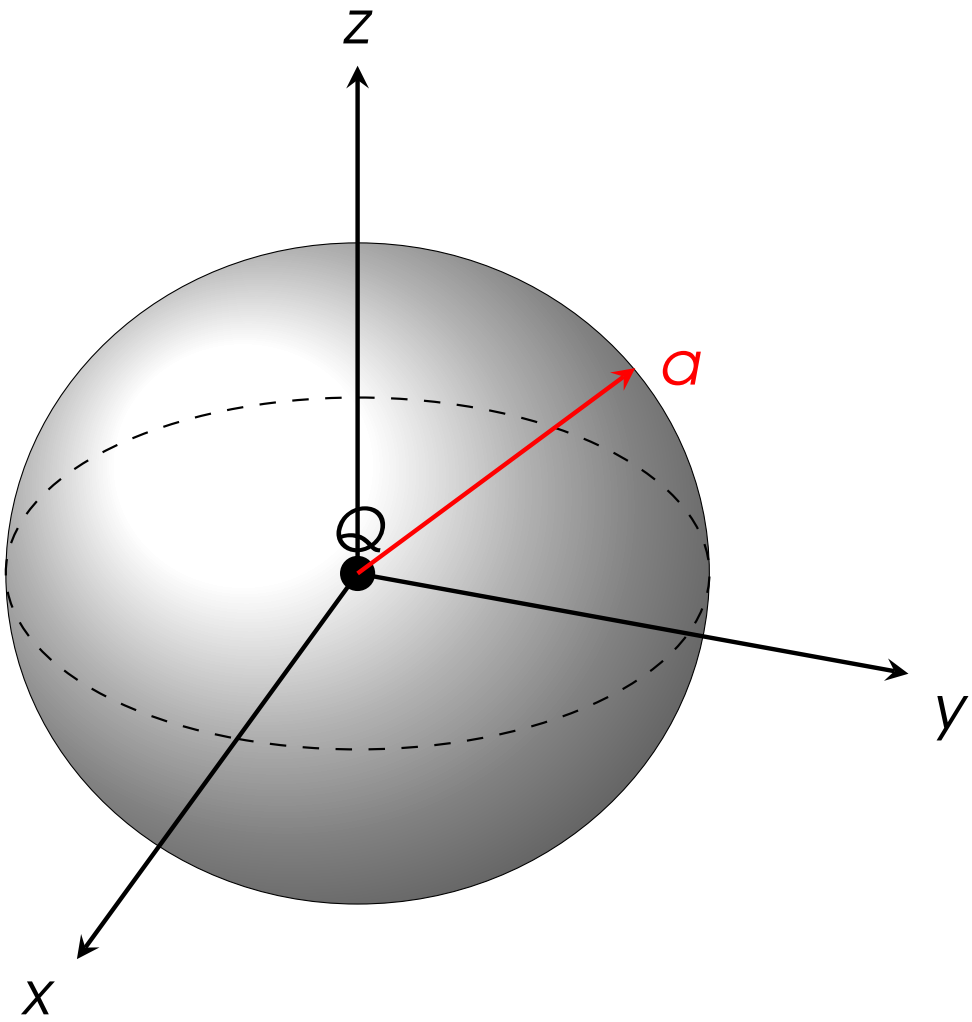
$\epsilon \vec{E}$

$\nabla \cdot \vec{E} = \frac{\rho_v}{\epsilon}$

$\oiint_S \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon}$

div. thm $\rightarrow$ $\iiint_V \nabla \cdot \vec{E}\, dV = \iiint_V \frac{\rho_v}{\epsilon}\, dV$

Gauss' Law: General Method



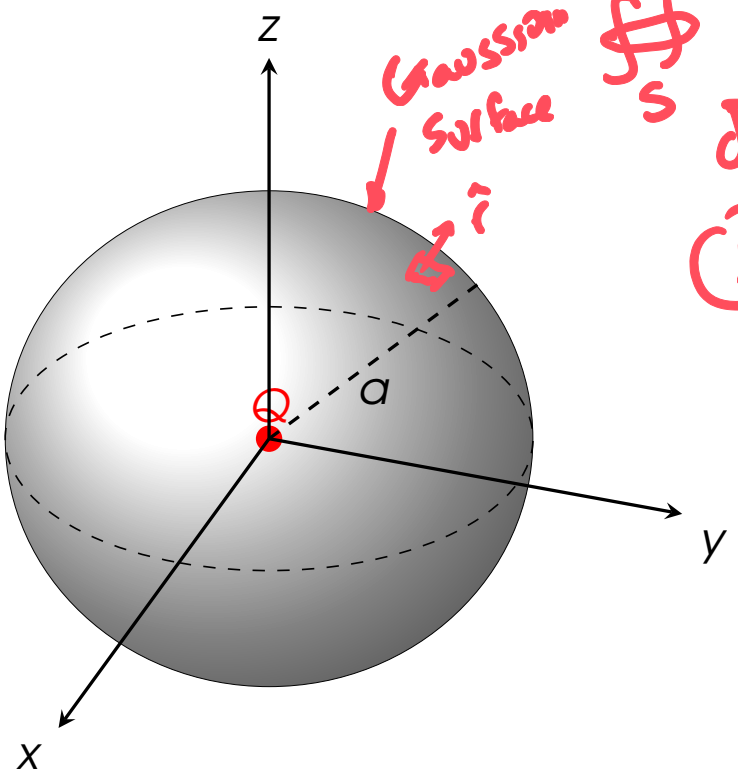
Method

1. Draw Gaussian Surface ($\vec{dS}$)
2. Determine symmetry
3. Solve Left Integral:

$$\oiint_S \varepsilon \vec{E} \cdot \vec{dS}$$

4. Determine charge enclosed, Q_{encl}

Gauss' Law: Point Charge



$$\oint_S \vec{E}\cdot\hat{n}\,dA = \frac{Q_{encl.}}{\epsilon} \quad (1)\ (2)$$

$$\vec{ds} = \hat{n}\,dA$$

$$(1)\quad \oint_S [\hat{r}E_r + \hat{\theta}E_\theta + \hat{\phi}E_\phi]\cdot\hat{r}\,r^2\sin\theta\,d\theta\,d\phi$$

$$\int_0^{2\pi}\int_0^{\pi} E_r\, r^2\sin\theta\,d\theta\,d\phi$$

E_r does not depend on θ or ϕ

$$\Rightarrow E_r\, r^2\, 4\pi$$

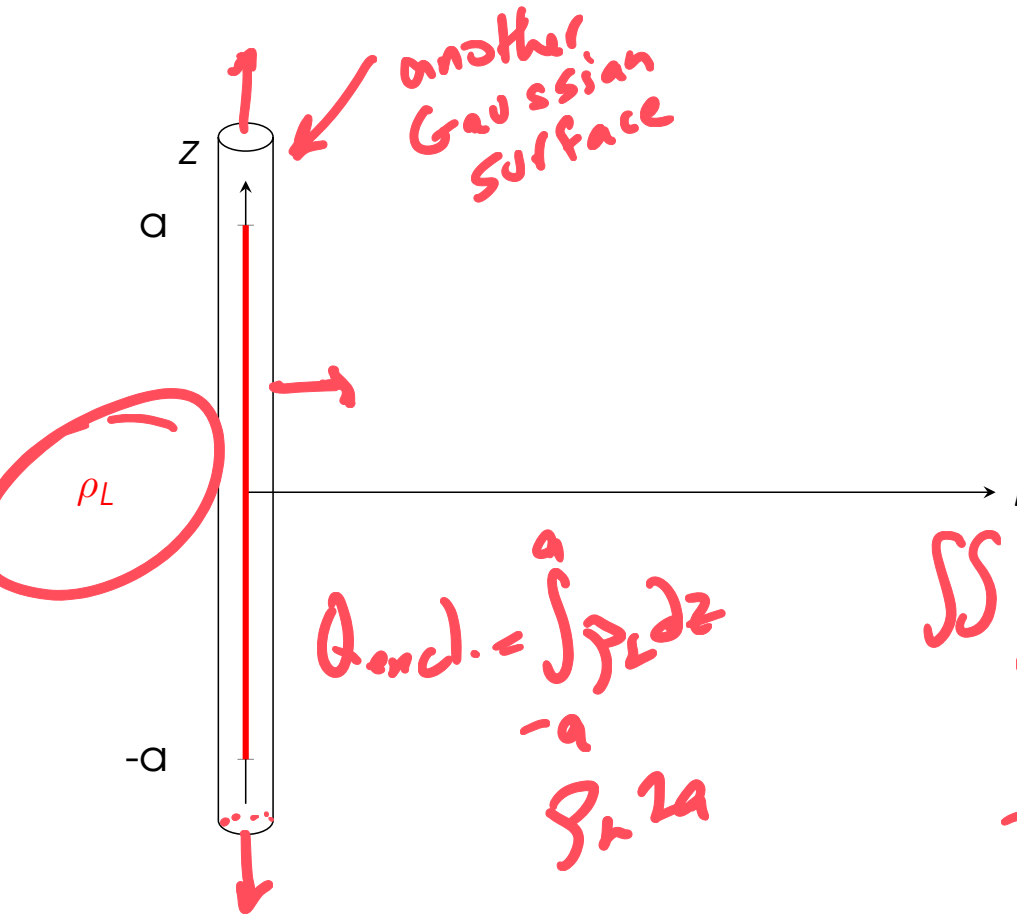
$$(1) = (2)$$

$$E_r\, r^2 4\pi = Q_{encl}/\epsilon$$

$$E_r = \frac{Q_{encl.}}{4\pi\epsilon r^2}$$

$$\vec{E} = \hat{r}E_r = \hat{r}\,\frac{Q_{encl.}}{4\pi\epsilon r^2}$$

Gauss' Law: Line Charge



$$\oint\oint \vec{E}\cdot d\vec{s} = \frac{Q_{encl.}}{\epsilon}$$

$$\int_{Top}\oint \vec{E}\cdot d\vec{s} + \int_{bottom}\oint \vec{E}\cdot d\vec{s} + \int_{side}\oint \vec{E}\cdot d\vec{s} = \frac{Q_{encl}}{\epsilon}$$

radius of G.S. $\to 0$

$$Q_{encl.} = \int_{-a}^{a} \rho_L\, dz$$

$$\rho_L 2a$$

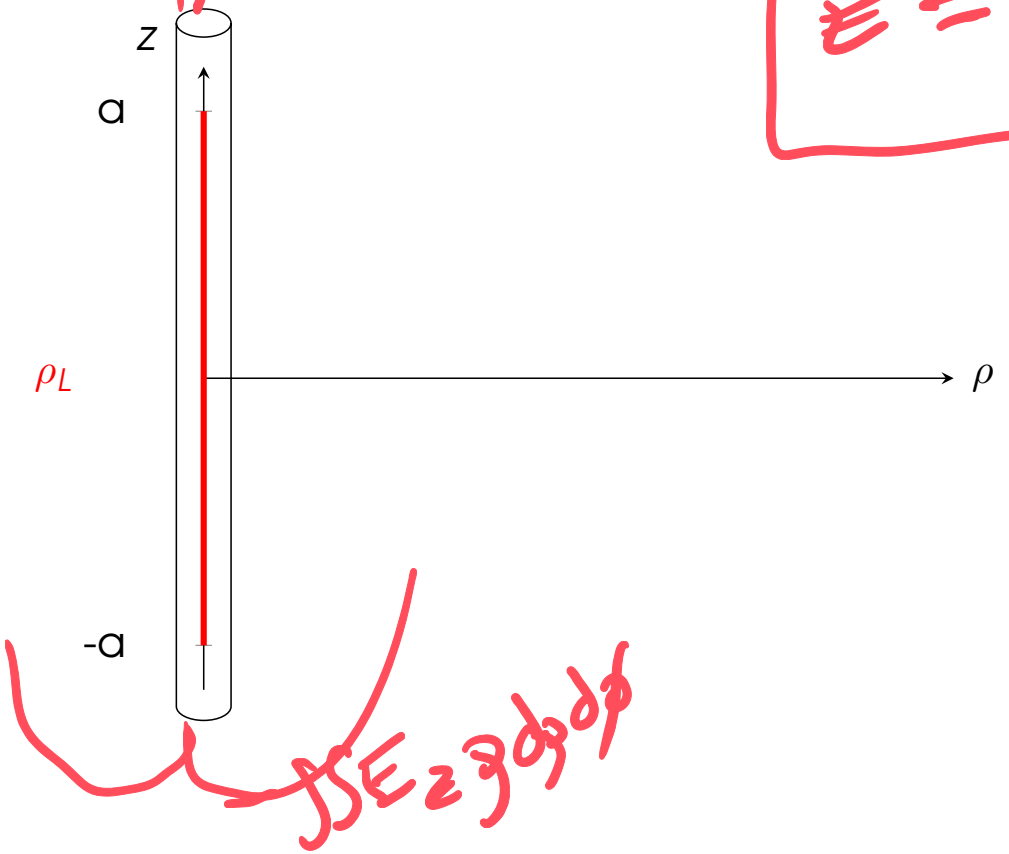
$$\int_{-a}^{a}\int_{0}^{2\pi}\left[\hat{\rho}E_\rho + \hat{\phi}E_\phi + \hat{z}E_z\right]\cdot\hat{\rho}\,\rho\, d\phi\, dz = \frac{Q_{encl.}}{\epsilon}$$

$$\int_{-a}^{a}\int_{0}^{2\pi} E_\rho\,\rho\, d\phi\, dz = \frac{Q_{encl.}}{\epsilon}$$

$$E_\rho\,\rho(2\pi)(2a) = \frac{Q_{encl.}}{\epsilon}$$

$$E_\rho = \frac{Q_{encl}}{4\pi\rho a\epsilon} = \frac{\rho_L 2a}{4\pi\rho a\epsilon}$$

Gauss' Law: Line Charge (cont'd)

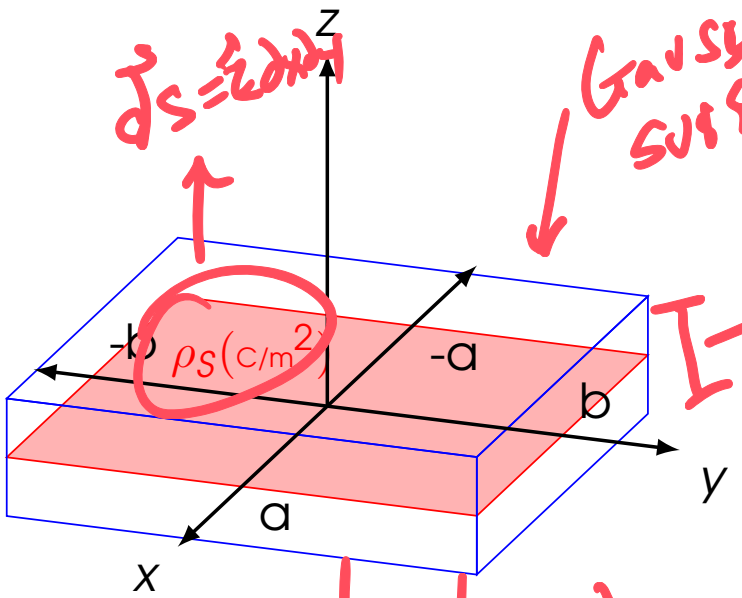


$$\vec{E} = \hat{\rho} E_\rho = \frac{\rho_L}{2\pi\epsilon\rho}$$

↳ Same as w/ Coulomb!!

Gauss' Law: Sheet Charge

Find the Electric field everywhere from a sheet carrying a charge (Uniform) distribution of ρ_S.



$\oint\oint \vec{E}\cdot d\vec{s} = \frac{Q_{encl.}}{\epsilon}$

$\iint_{top} \vec{E}\cdot d\vec{s} + \iint_{bottom} \vec{E}\cdot d\vec{s} + \iint_{sides} \vec{E}\cdot d\vec{s} = \frac{Q_{encl}}{\epsilon}$

$TOP = \int_{-b}^{b}\int_{-a}^{a} E_z\, dx\, dy = 4ab E_z$

$Bottom = \int_{-b}^{b}\int_{-a}^{a} (-E_z)(-\hat{z}\, dx\, dy) = 4ab E_z$

$2(4ab)E_z = \frac{Q_{encl.}}{\epsilon}$

$= \frac{1}{\epsilon}\int_{-b}^{b}\int_{-a}^{a} \rho_s\, dx\, dy$

$= 4ab\,\rho_s/\epsilon$

Gaussian surface

$I \to 0$

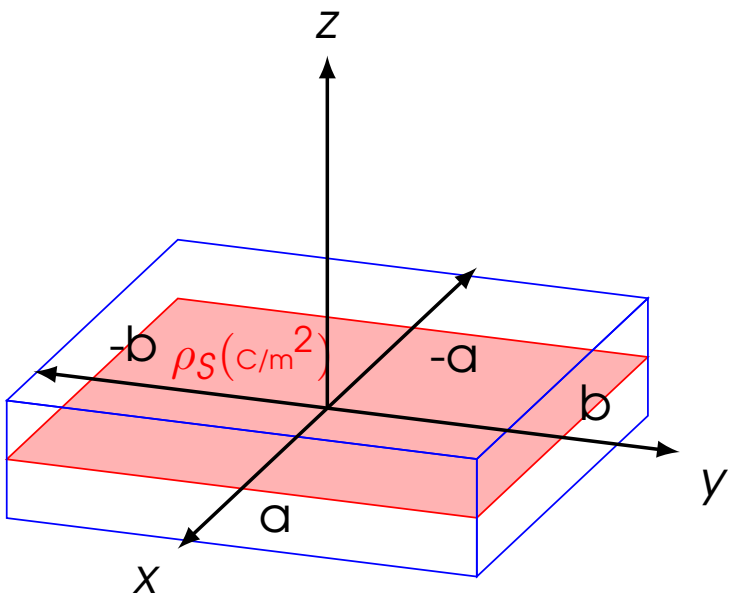
$\vec{ds} = \hat{z}\, dx\, dy$

$-\hat{z}E_z$ $-\hat{z}\, dx\, dy$

$\vec{E} = \hat{z}\frac{\rho_s}{2\epsilon}$

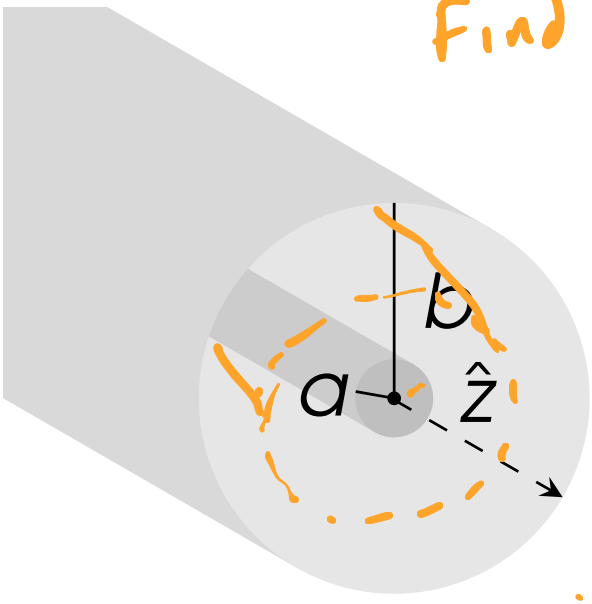
Gauss' Law: Sheet Charge (cont'd)

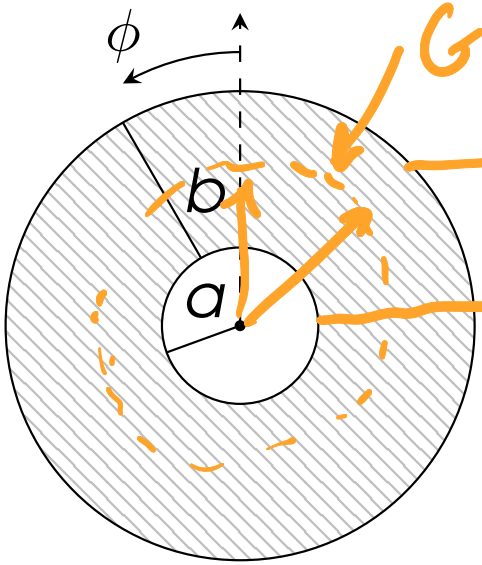
Find the Electric field everywhere from a sheet carrying a charge distribution of ρ_S.



Gauss' Law: Coaxial Cable

Find $\vec{E}$ everywhere...





Gaussian Surface

$+ V_0$

1) Draw a picture ✓

2) With Gauss' Law:

$$\oint_S \vec{E} \cdot d\vec{s} = Q_{encl.}/\epsilon$$

(a) (b)

(a) $\vec{E} = \hat{\rho} E_\rho + \hat{\phi} E_\phi + \hat{z} E_z$

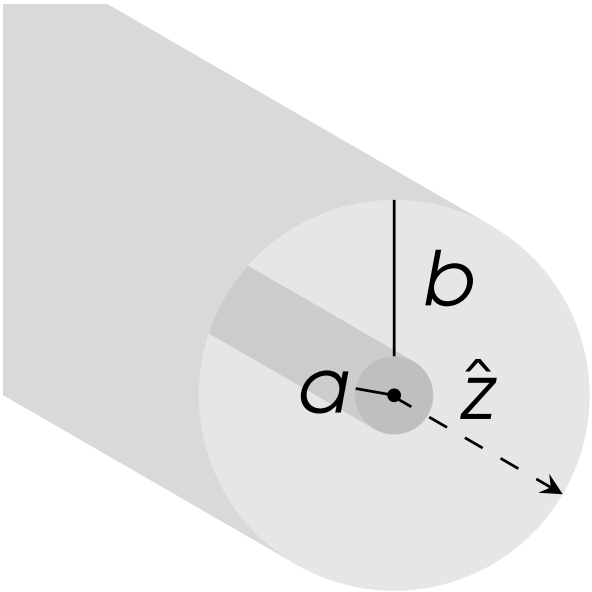
$d\vec{s} = \hat{\rho}\, \rho\, d\phi\, dz$

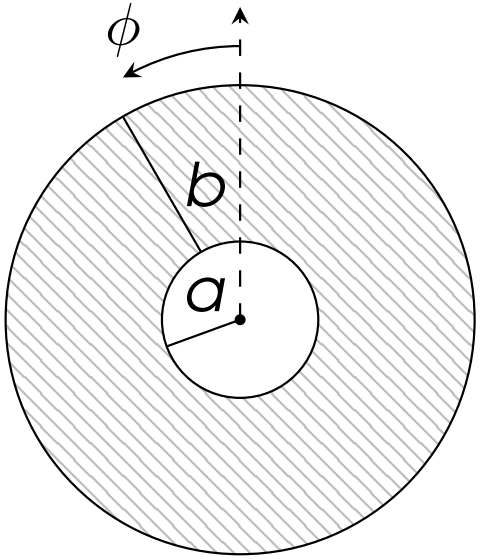
$$\int_0^{2\pi}\int_0^{L} E_\rho\, \rho\, d\phi\, dz$$

$2\pi E_\rho \rho L$

(b)

Gauss' Law: Coaxial Cable (cont'd)





(b) $Q_{encl.} = \iiint \rho_v \, dV$

$\Rightarrow \rho_v = \frac{Q_{TOT}}{\pi a^2 L}$

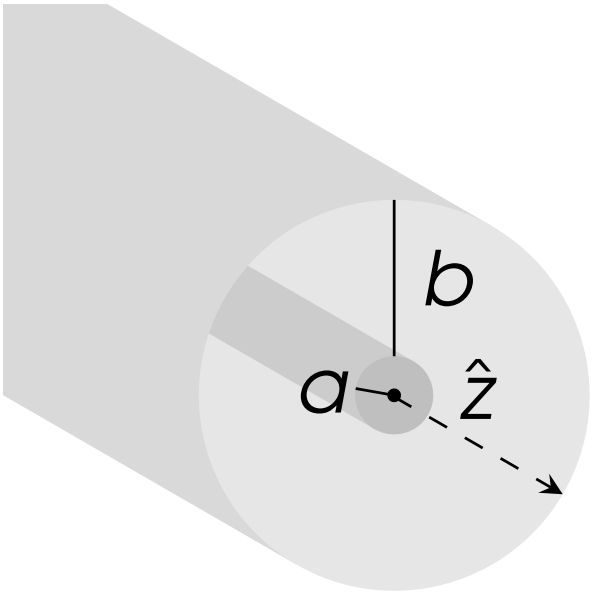
$Q_{encl.} = \int_0^L \int_0^{2\pi} \int_0^{\rho_0} \frac{Q_{TOT}}{\pi a^2 L} \rho \, d\rho \, d\phi \, dz$

$= \frac{Q_{TOT}}{\pi a^2 L} \int_0^L dz \int_0^{2\pi} d\phi \int_0^{\rho_0} \rho \, d\rho$

$= \frac{Q_{TOT}}{\pi a^2 L} (L)(2\pi)\left(\frac{\rho_0^2}{2}\right)$

$Q_{encl} = \frac{Q_{TOT}\,\rho_0^2}{a^2}$

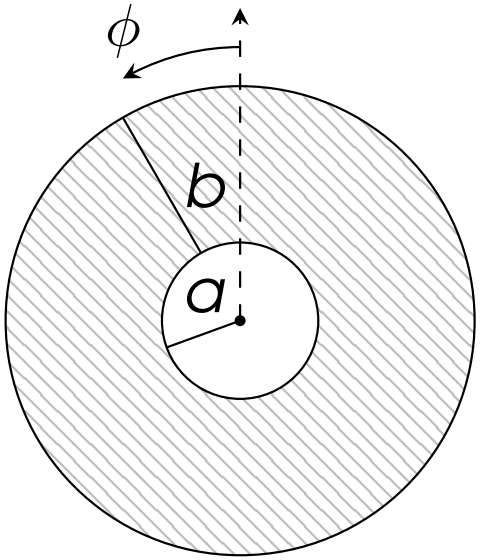
Gauss' Law: Coaxial Cable (cont'd)



$$2\pi E_\rho \rho L = \left(\frac{1}{\epsilon}\right)\left(\frac{Q_{tot}\rho^2}{a^2}\right)$$

$$\boxed{E_\rho = \frac{Q_{tot}\rho^2}{\epsilon a^2 \rho 2\pi L}}$$

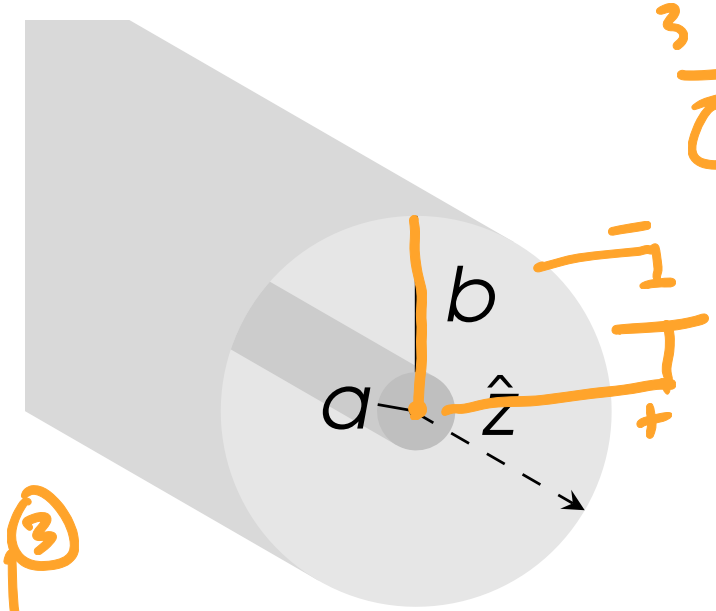
$$\vec{E} = \hat{\rho} E_\rho$$



$\rho \rightarrow$ Observation radius $\rho_0 \rightarrow$ "source radius"

Gauss' Law: Coaxial Cable (cont'd)

$$E_\rho = \frac{Q_{TOT}\,\rho_0^2}{2\pi a^2 \epsilon L \rho}$$



3 cases:

① $\rho < a \Rightarrow \rho_0 = \rho$

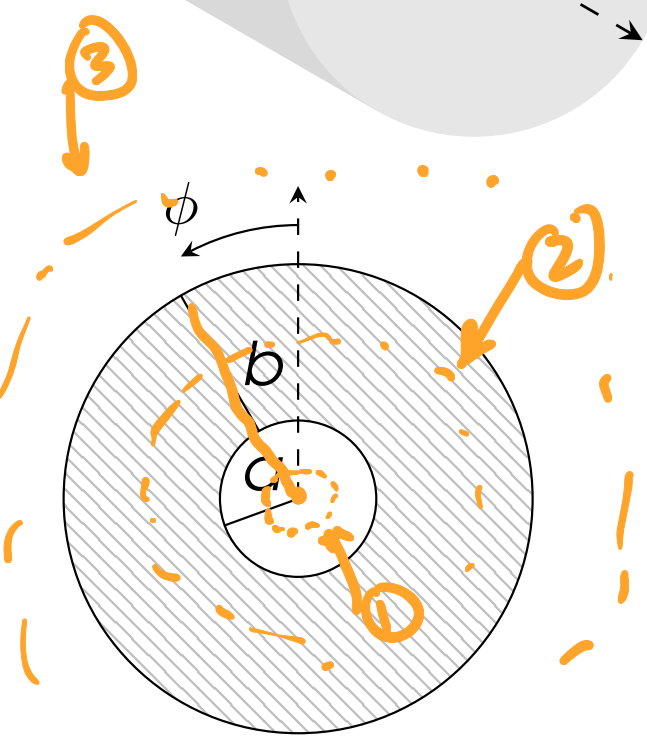
$$E_\rho = \frac{Q_{TOT}\,\rho}{2\pi a^2 \epsilon L \rho} = \boxed{\frac{Q_{TOT}}{2\pi a^2 \epsilon L}}$$ ✓

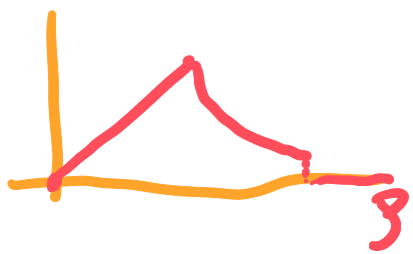
② $a < \rho < b \Rightarrow \rho_0 = a$

$$E_\rho = \frac{Q_{TOT}\,a}{2\pi a^2 \epsilon L \rho} = \frac{Q_{TOT}}{2\pi \epsilon L \rho}$$ ✓

③ $\rho > b$ $Q_{TOT} = +q - q = 0$

$E_\rho = 0$





Gauss' Law: Point (differential) Form

Derivation

$$\oiint \vec{D} \cdot d\vec{S} = Q_{\text{encl}}$$

$$\iiint_V \nabla \cdot \vec{D} dV = \iiint_V \rho_V dV$$

$$\nabla \cdot \vec{D} = \rho_V$$

$$\nabla \cdot \vec{E} = \frac{\rho_V}{\varepsilon}$$

EXAMPLE

Given the following static electric field in free space, find the charge density (ρ) at the point $(2, -2, 0)$

$$\vec{E}(\vec{r}) = \hat{\mathbf{x}}Ax^2 + \hat{\mathbf{y}}Bz + \hat{\mathbf{z}}Cx^2z \longrightarrow A = 3\text{V}/\text{m}^3 \quad B = 2\text{V}/\text{m}^3 \quad C = 1\text{V}/\text{m}^4$$

Solution

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

$$\varepsilon_0 \left[\frac{\partial}{\partial x}\left(3x^2\right) + \frac{\partial}{\partial y}\left(2z\right) + \frac{\partial}{\partial z}\left(x^2z\right) \right] = \rho$$

$$\varepsilon_0 \left(6x + 0 + x^2\right) = \rho$$

$$8.854 \times 10^{-12}\text{F/m}\left[(6)(2) + 2^2\right] = \rho$$

$$141.7\text{pC/m}^3 = \rho$$